Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Features tracked

37

Missing Values

In review ratings

Customer Data

- Demographics (Age, Gender, Location)
- Subscription Status
- Review Ratings

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Discount & Promo Codes

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Data Preparation Process

Data Loading

Imported dataset using pandas

Initial Exploration

Checked structure and summary statistics

Missing Data Handling

Imputed missing values using median ratings

Standardization

Renamed columns to snake case

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Feature Engineering

Age Binning

Created age categories for segmentation

Purchase Frequency

Calculated days between purchases

Data Consistency

Dropped redundant promo code column

MySQL Integration

Loaded cleaned data for SQL analysis

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SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue generated by male vs. female customers

High-Spending Discount Users

839 customers used discount but spent above average purchase amount

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Product Performance Analysis

1

Top 5 Products by Rating

Identified highest average review ratings

2

Discount-Dependent Products

5 products with highest percentage of discounted purchases

3

Top 3 Products per Category

Listed most purchased products within each category

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Shipping & Subscription Analysis

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping

Subscribers vs Non-Subscribers

Compared average spend and total revenue across subscription status

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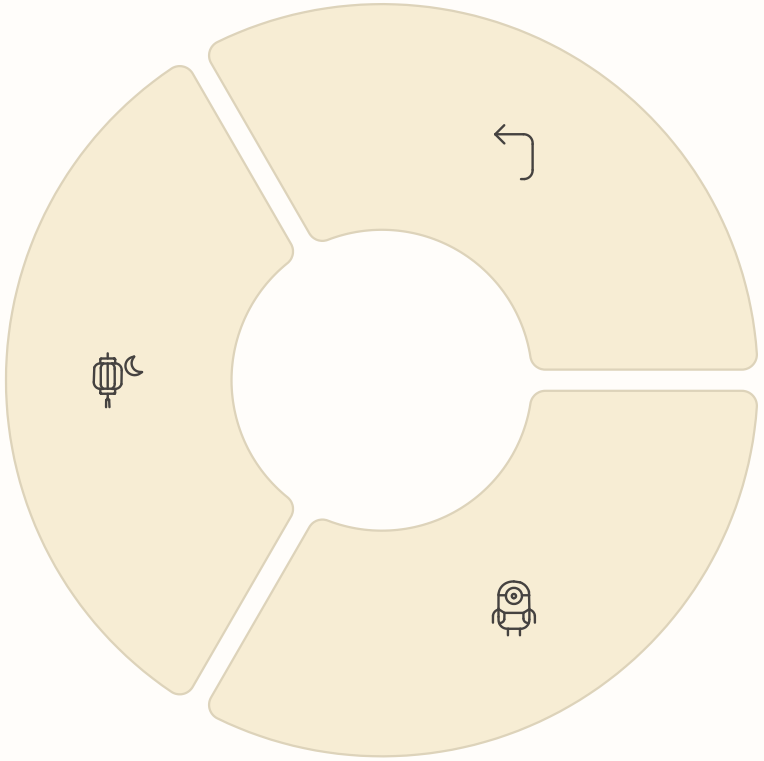
Customer Segmentation



New Customers

First-time purchases

Returning Customers

Multiple purchases

Loyal Customers

Consistent purchase history

Key Finding: Customers with >5 purchases are more likely to subscribe

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Revenue by Age Group

Calculated total revenue contribution of each age group

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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